



Predictive Value of C-Reactive Protein, Procalcitonin, and Blood Lactate for Septic Shock in Children With Sepsis in Northern Mountainous Vietnam

Duy Thai Khang Tran¹, Thanh Trung Nguyen^{2,3*}, Cong Tien Nguyen⁴, Đức Duy Lê⁵

¹ Department of Pediatrics, Da Nang Family Hospital, 550000 Da Nang City, Vietnam

² VNU University of Medicine and Pharmacy, Vietnam National University, 100000 Hanoi, Vietnam

³ Department of Tropical Diseases, Dien Bien Provincial General Hospital, 380000 Dien Bien Phu, Vietnam

⁴ Department of Pediatrics, Hoan My Da Nang Hospital, 550000 Da Nang City, Vietnam

⁵ Department of Training Management, Institute for Research and Training in Community Healthcare, 100000 Hanoi, Vietnam

* Correspondence: Thanh Trung Nguyen (nguyen.hnvnu@gmail.com)

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Abstract: Early identification of septic shock in children with sepsis remains challenging, particularly in resource-limited and geographically disadvantaged settings. The predictive performance of C-reactive protein, procalcitonin, and blood lactate for septic shock was evaluated in a multicenter retrospective study involving 245 children aged <10 years who were diagnosed with sepsis at six provincial hospitals in northern mountainous Vietnam between 2022 and 2025. Patients were classified into septic shock ($n = 82$) and non-septic shock ($n = 163$) groups. Clinical characteristics and C-reactive protein, procalcitonin, and blood lactate concentrations measured within 24 hours of hospital admission were evaluated. Biomarker discrimination was assessed using receiver operating characteristic curve analysis. The median age of the study population was 35 months (interquartile range, 18.0–54.0 months), and septic shock was identified in 82 children (33.5%). Fever $>38.5^{\circ}\text{C}$ (adjusted odds ratio, 3.91; 95% confidence interval, 1.62–9.43), cardiovascular dysfunction (adjusted odds ratio, 10.94; 95% confidence interval, 5.19–23.08), and neurological dysfunction (adjusted odds ratio, 6.07; 95% confidence interval, 2.80–13.16) were independently associated with septic shock. Among the three biomarkers evaluated, blood lactate showed the greatest discriminatory ability, with an area under the receiver operating characteristic curve (AUC) of 0.713 (95% confidence interval, 0.640–0.785) and an optimal cutoff of 3.93 mmol/L. Procalcitonin showed lower discriminatory performance (AUC, 0.652; 95% confidence interval, 0.577–0.727), whereas C-reactive protein did not discriminate between patients with and without septic shock (AUC, 0.485; $p = 0.711$). These findings suggest that, among the biomarkers evaluated, blood lactate provides the greatest discriminatory value for septic shock in children with sepsis. Procalcitonin may provide complementary information, whereas C-reactive protein appears to have limited discriminatory utility for this purpose. Blood lactate, potentially supplemented by procalcitonin and readily assessed clinical indicators, may therefore contribute to early risk stratification for septic shock in pediatric sepsis, particularly in resource-limited healthcare settings.

Keywords: Sepsis; Septic shock; Children; C-reactive protein; Procalcitonin; Blood lactate; Northern mountainous Vietnam

1. Introduction

Sepsis is one of the leading causes of mortality and disability among children worldwide (Fleischmann-Struzek et al., 2018; Weiss et al., 2020). According to the Global Burden of Disease (GBD) study, there were approximately 48.9 million cases of sepsis and 11 million sepsis-related deaths globally in 2017, with children under five years of age being the most severely affected population (Rudd et al., 2020). Septic shock is a particularly severe complication of sepsis, characterized by profound circulatory and metabolic abnormalities that may result in

multiple organ dysfunction and death if not recognized and treated promptly (Schlapbach et al., 2024; Singer et al., 2016; Weiss et al., 2020).

In clinical practice, early identification of septic shock in children remains challenging because of its heterogeneous clinical manifestations and rapid disease progression (Hartman et al., 2013; Schlapbach et al., 2024). Blood culture is still considered the gold standard for the diagnosis of sepsis owing to its ability to identify causative pathogens and determine antimicrobial susceptibility (Lamy et al., 2016). However, this method has several limitations, including a long turnaround time and a relatively low positivity rate, especially in patients who have received antibiotics before hospital admission (Coburn et al., 2012; Lamy et al., 2016). Therefore, biomarkers such as C-reactive protein, procalcitonin, and blood lactate have attracted increasing attention because of their potential roles in early diagnosis, assessment of disease severity, and prognostic evaluation (Faix, 2013; Lê et al., 2022; Pierrakos & Vincent, 2010; Simon et al., 2004).

Numerous studies worldwide have demonstrated that procalcitonin and blood lactate are valuable biomarkers for predicting septic shock in children, whereas the predictive role of C-reactive protein remains controversial. Blood lactate is considered an indicator of tissue hypoperfusion and metabolic dysfunction and has been shown to be closely associated with disease severity and mortality risk in patients with septic shock (Fernando et al., 2018; Jat et al., 2011). Procalcitonin is a more specific biomarker for bacterial infection and has demonstrated high sensitivity and specificity in both the diagnosis and prognostic assessment of sepsis (Lê et al., 2022; Meisner, 2014; Simon et al., 2004). In contrast, the predictive value of C-reactive protein for septic shock varies across studies and may be influenced by factors such as the timing of disease onset, the magnitude of the inflammatory response, and differences in study populations (Lelubre et al., 2013; Póvoa et al., 2005). Nevertheless, most existing studies have been conducted in large medical centers or developed countries, and evidence from mountainous, remote, and underserved regions remains limited.

The northern mountainous provinces of Vietnam are characterized by complex terrain, fragmented mountainous landscapes, difficult transportation, and uneven population distribution. This region is home to a large proportion of ethnic minority populations, including the Hmong, Tay, Dao, Nung, and other ethnic groups. Children from ethnic minority communities often face substantial barriers to accessing healthcare services, including long travel distances, limited economic resources, language barriers, and differences in health-seeking behaviors (Målqvist et al., 2013). These challenges may result in delayed hospital admission and consequently increase the risk of severe disease progression and septic shock. In this current study, more than half of the patients belonged to ethnic minority groups, reflecting the demographic characteristics of the study area. However, data regarding the predictive value of biomarkers for septic shock in this specific population remain scarce (Rudd et al., 2018).

Given these considerations, the present study was conducted to evaluate the predictive value of C-reactive protein, procalcitonin, and blood lactate for septic shock in children with sepsis treated at hospitals in the northern mountainous provinces of Vietnam. The findings of this study are expected not only to provide additional evidence regarding the prognostic utility of these biomarkers but also to have practical implications for the early identification, risk stratification, and optimization of treatment strategies for pediatric sepsis in mountainous areas, where the proportion of ethnic minority patients is high and access to healthcare remains limited.

2. Methodology

2.1 Study Design and Participants

A multicenter retrospective study was conducted on pediatric patients with sepsis who were treated at hospitals in the northern mountainous provinces of Vietnam between January 2022 and September 2025. The study was performed at six provincial general hospitals located in Dien Bien, Thai Nguyen, Lao Cai, Cao Bang, Bac Kan, and Yen Bai provinces. A total of 245 children younger than 10 years of age were enrolled in the study. The inclusion criteria were as follows: (i) children diagnosed with sepsis according to current international criteria; (ii) availability of complete measurements of C-reactive protein, procalcitonin, and blood lactate obtained within the first 24 hours after hospital admission; and (iii) availability of complete clinical information regarding patient characteristics, septic shock status, and clinical outcomes. The exclusion criteria included (i) missing data on C-reactive protein, procalcitonin, or blood lactate; (ii) incomplete medical records or inability to determine clinical outcomes; and (iii) severe chronic diseases or metabolic disorders that could substantially affect the levels of the investigated biomarkers. Based on the presence or absence of septic shock, patients were classified into two groups: the septic shock group and the non-septic shock group. Septic shock was defined according to current pediatric guidelines as sepsis accompanied by severe circulatory and metabolic abnormalities, requiring vasoactive agents to maintain age-appropriate blood pressure or presenting with signs of tissue hypoperfusion associated with elevated blood lactate levels.

2.2 Data Collection and Study Variables

Study data were retrospectively collected from electronic and paper-based medical records of patients who met

the eligibility criteria. Information recorded included demographic characteristics, clinical manifestations, and laboratory findings obtained at hospital admission or within the first 24 hours after admission. C-reactive protein, procalcitonin, and blood lactate measurements were obtained as part of routine clinical assessment at the participating hospitals. For each biomarker, the first available measurement obtained after hospital admission and within the first 24 hours was used for analysis. When multiple measurements were available during this period, only the earliest value was included; peak or subsequent values were not used. Thus, biomarker selection was based on the timing of hospital admission rather than on the timing of septic shock diagnosis. Laboratory measurements at all six participating hospitals were performed in accordance with standardized laboratory procedures and quality-control requirements applicable in Vietnam. However, because of the multicenter retrospective design, the specific analyzers, assay platforms, reagent systems, and site-specific reference ranges were not necessarily identical across the participating hospitals. Each laboratory performed the assays according to its established standard operating procedures and the manufacturers' instructions, with routine internal quality control. C-reactive protein concentrations were expressed in mg/L, procalcitonin in ng/mL, and blood lactate in mmol/L. All data were anonymized, coded, and entered into a dedicated study database before statistical analysis to ensure data accuracy and patient confidentiality.

2.3 Diagnostic Criteria for Septic Shock

Sepsis was diagnosed according to current international pediatric criteria and was defined as suspected or confirmed infection accompanied by organ dysfunction resulting from a dysregulated host response to infection (Miranda & Nadel, 2023). Septic shock was defined as sepsis associated with profound circulatory and metabolic abnormalities, manifested by persistent hypotension or signs of tissue hypoperfusion requiring vasoactive agents to maintain age-appropriate blood pressure (Weiss et al., 2020). According to the 2020 Surviving Sepsis Campaign guidelines, pediatric septic shock is characterized by severe infection leading to cardiovascular dysfunction, including hypotension, the need for vasoactive support, or evidence of impaired tissue perfusion (Weiss et al., 2020).

For the present retrospective study, septic shock status was determined from the clinical diagnosis and hemodynamic findings documented in the medical records. Patients were classified into the septic shock group when sepsis was accompanied by persistent hypotension and/or cardiovascular dysfunction requiring vasoactive support to maintain age-appropriate blood pressure despite adequate fluid resuscitation, or by documented clinical evidence of impaired tissue perfusion. Clinical indicators of impaired perfusion available in the medical records included abnormal capillary refill, weak peripheral pulses, altered mental status, and/or reduced urine output. Patients with sepsis who did not meet these criteria were classified into the non-septic shock group. Importantly, blood lactate concentration was not used as a criterion for assigning patients to the septic shock or non-septic shock groups. Septic shock status was used as the primary outcome variable to evaluate the predictive value of the investigated biomarkers, including C-reactive protein, procalcitonin, and blood lactate.

2.4 Statistical Analysis

Data were entered and analyzed using SPSS version 22.0 (IBM Corp., Armonk, NY, USA). Differences between the septic shock and non-septic shock groups were assessed using the chi-square test or Fisher's exact test, as appropriate. Univariate logistic regression analysis was initially performed to identify factors associated with septic shock. Variables with a p -value < 0.10 in the univariate analysis were subsequently included in the multivariate logistic regression model to determine independent predictors. The results were expressed as odds ratios or adjusted odds ratios with corresponding 95% confidence intervals. The predictive performance of C-reactive protein, procalcitonin, and blood lactate for septic shock was evaluated using receiver operating characteristic curve analysis. The area under the receiver operating characteristic curve (AUC), sensitivity, specificity, and optimal cutoff values were determined based on the Youden index. A two-sided p -value < 0.05 was considered statistically significant.

3. Results

3.1 General Characteristics of Pediatric Patients With Sepsis

The general characteristics of the 245 pediatric patients with sepsis included in this study are presented in Table 1. The median age of the patients was 35 months (interquartile range, 18.0–54.0 months), indicating that sepsis in the study population occurred predominantly in children younger than 5 years of age. Male patients slightly outnumbered females, accounting for 53.5% and 46.5% of the study population, respectively. Regarding ethnicity, ethnic minority children accounted for 58.0% of all participants. A total of 45.8% of the patients were malnourished to varying degrees, including mild malnutrition (22.9%), moderate malnutrition (14.3%), and severe

malnutrition (8.6%). The median length of hospital stay was 11 days (interquartile range, 8–13 days). Among the 245 patients enrolled, 82 cases (33.5%) developed septic shock.

Table 1. General characteristics of pediatric patients with sepsis enrolled in the study ($n = 245$)

Characteristic		Value
Age (months), median (interquartile range)		35.0 (18.0–54.0)
Sex, n (%)	Male	131 (53.5)
	Female	114 (46.5)
Ethnicity, n (%)	Kinh	103 (42.0)
	Hmong	45 (18.4)
	Tay	39 (15.9)
	Dao	34 (13.9)
	Nung	24 (9.8)
	Normal	133 (54.3)
Nutritional status, n (%)	Mild malnutrition	56 (22.9)
	Moderate malnutrition	35 (14.3)
	Severe malnutrition	21 (8.6)
Length of hospital stay (days), median (interquartile range)		11 (8–13)
Septic shock, n (%)		82 (33.5)

3.2 Clinical Characteristics Associated With Septic Shock in Children With Sepsis

The clinical characteristics associated with septic shock are presented in Table 2. Overall, all clinical manifestations were more frequently observed in patients with septic shock than in those without shock. Fever >38.5 °C was more frequently observed in patients with septic shock than in those without septic shock (85.4% vs. 71.2%; $\chi^2 = 6.017$, $p = 0.014$). Similarly, respiratory dysfunction was more frequent in the septic shock group than in the non-septic shock group (57.3% vs. 37.4%; $\chi^2 = 8.759$, $p = 0.003$). Cardiovascular dysfunction (59.8% vs. 11.7%; $\chi^2 = 62.946$, $p < 0.001$) and neurological dysfunction (46.3% vs. 12.9%; $\chi^2 = 33.404$, $p < 0.001$) showed particularly strong associations with septic shock. Gastrointestinal dysfunction was also more frequent in patients with septic shock than in those without septic shock (40.2% vs. 27.6%; $\chi^2 = 4.014$, $p = 0.045$).

Table 2. Comparison of clinical characteristics between the septic shock and non-septic shock groups

Clinical Characteristic	Non-Septic Shock ($n = 163$), x (%)	Septic Shock ($n = 82$), x (%)	χ^2	p -Value
Fever >38.5 °C	116 (71.2%)	70 (85.4%)	6.017	0.014
Respiratory dysfunction	61 (37.4%)	47 (57.3%)	8.759	0.003
Cardiovascular dysfunction	19 (11.7%)	49 (59.8%)	62.946	<0.001
Neurological dysfunction	21 (12.9%)	38 (46.3%)	33.404	<0.001
Gastrointestinal dysfunction	45 (27.6%)	33 (40.2%)	4.014	0.045

Note: Data are presented as x (%), with percentages calculated within each septic shock status group.

Multivariate logistic regression analysis showed that fever >38.5 °C, cardiovascular dysfunction, and neurological dysfunction were independently associated with septic shock in children with sepsis (Table 3). Specifically, patients with fever >38.5 °C had a 3.91-fold higher risk of developing septic shock compared with those without high fever (adjusted odds ratio, 3.91; 95% confidence interval, 1.62–9.43; $p = 0.002$). Cardiovascular dysfunction was the strongest predictor, increasing the risk of septic shock by 10.94-fold (adjusted odds ratio, 10.94; 95% confidence interval, 5.19–23.08; $p < 0.001$), followed by neurological dysfunction, which was associated with a 6.07-fold increased risk (adjusted odds ratio, 6.07; 95% confidence interval, 2.80–13.16; $p < 0.001$). In contrast, respiratory dysfunction (adjusted odds ratio, 1.51; $p = 0.234$) and gastrointestinal dysfunction (adjusted odds ratio, 1.71; $p = 0.145$) were not significantly associated with septic shock after adjustment for other variables in the multivariate model.

3.3 Predictive Value of C-Reactive Protein, Procalcitonin, and Blood Lactate for Septic Shock in Children With Sepsis

The receiver operating characteristic curves demonstrated that blood lactate was located farther from the reference diagonal line and above the curves of procalcitonin and C-reactive protein, indicating superior predictive performance for septic shock among the three biomarkers evaluated. As shown in Figure 1 and Table 4, blood lactate exhibited the best predictive value, with an AUC of 0.713 (95% confidence interval, 0.640–0.785; $p < 0.001$), indicating an acceptable discriminatory ability. The optimal cutoff value for blood lactate was 3.93 mmol/L, yielding a sensitivity of 50.0% and a specificity of 87.1%. Procalcitonin also demonstrated predictive value for

septic shock, with an AUC of 0.652 (95% confidence interval, 0.577–0.727; $p < 0.001$). The optimal cutoff value was 10.05 ng/mL, corresponding to a sensitivity of 52.4% and a specificity of 73.0%. In contrast, C-reactive protein showed no significant predictive value for septic shock in this study, with an AUC of 0.485 (95% confidence interval, 0.407–0.564; $p = 0.711$). Overall, procalcitonin showed moderate predictive performance, whereas the receiver operating characteristic curve of C-reactive protein was located close to the reference diagonal line, indicating poor discriminative ability. These findings suggest that blood lactate and procalcitonin may serve as useful biomarkers for the early identification of children with sepsis who are at risk of developing septic shock in the northern mountainous provinces of Vietnam.

Table 3. Multivariate logistic regression analysis of factors associated with septic shock in children with sepsis

Clinical Characteristic	Adjusted Odds Ratio	95% Confidence Interval	<i>p</i> -Value
Fever >38.5 °C	3.91	1.62–9.43	0.002
Respiratory dysfunction	1.51	0.77–2.98	0.234
Cardiovascular dysfunction	10.94	5.19–23.08	<0.001
Neurological dysfunction	6.07	2.80–13.16	<0.001
Gastrointestinal dysfunction	1.71	0.83–3.51	0.145

Table 4. Predictive value of C-reactive protein, procalcitonin, and blood lactate for septic shock in children with sepsis

Biomarker	Area Under the Receiver Operating Characteristic Curve (AUC)	95% Confidence Interval	Cutoff Value	Sensitivity (%)	Specificity (%)	<i>p</i> -Value
Procalcitonin (ng/mL)	0.652	0.577–0.727	10.05	52.4	73.0	<0.001
C-reactive protein (mg/L)	0.485	0.407–0.564	85.30	50.0	50.3	0.711
Blood lactate (mmol/L)	0.713	0.640–0.785	3.93	50.0	87.1	<0.001

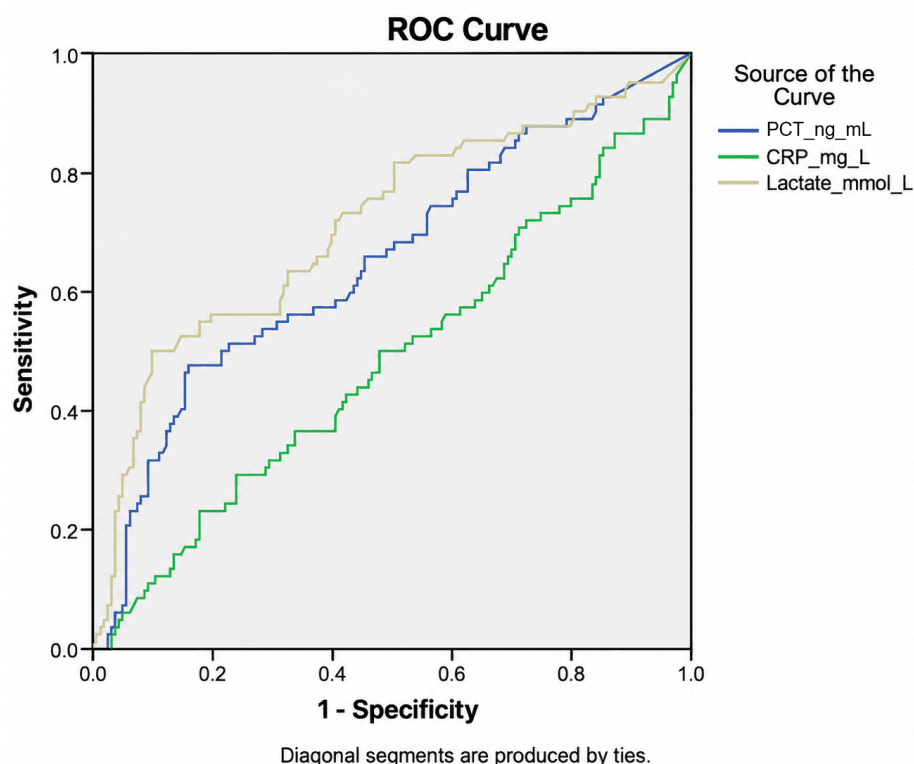


Figure 1. Receiver operating characteristic curves of C-reactive protein, procalcitonin, and blood lactate for predicting septic shock in children with sepsis

Note: ROC = receiver operating characteristic; PCT_ng_mL = Procalcitonin (ng/mL); CRP_mg_L = C-reactive protein (mg/L); Lactate_mmol_L = Blood lactate (mmol/L).

4. Discussion

The median age of patients in the present study was 35 months, indicating that sepsis occurred predominantly in children younger than 5 years of age. This finding is consistent with the study by Rudd et al. (2020), which reported that children under 5 years, particularly neonates and infants, represent the age group most frequently affected by sepsis. Male patients accounted for 53.5% of the study population, slightly higher than females. This finding differs from previous reports showing that females had a higher age-standardized incidence of sepsis than males (716.5 vs. 642.8 cases per 100,000 population), although male patients tended to experience more severe disease (Rudd et al., 2020). In addition, the prevalence of septic shock among children with sepsis in the current study was 33.5%. This relatively high proportion reflects the substantial burden of septic shock in the northern mountainous provinces of Vietnam. The findings of this current study are comparable to those reported by Schlappbach et al. (2024), in which the prevalence of pediatric septic shock ranged from 20% to 40%. The high prevalence of septic shock observed in the current study may be partly explained by the characteristics of the study region, where the majority of patients belonged to ethnic minority groups (58%). Children from ethnic minority communities often face barriers to healthcare access, including long travel distances, limited economic resources, language differences, and disparities in health-seeking behaviors, which may result in delayed hospital admission and more severe disease progression (Målqvist et al., 2013).

Furthermore, the high prevalence of malnutrition (45.8%) observed in the current study may have contributed to the increased risk of severe disease and septic shock. Therefore, the high proportion of ethnic minority patients not only reflects the demographic characteristics of the study region but also highlights the need to prioritize these vulnerable populations in programs aimed at the early detection and management of pediatric sepsis in Vietnam. The current study demonstrated that clinical manifestations such as fever $>38.5^{\circ}\text{C}$, respiratory dysfunction, cardiovascular dysfunction, neurological dysfunction, and gastrointestinal dysfunction were all more frequently observed in patients with septic shock than in those without shock. Among these manifestations, cardiovascular and neurological dysfunction showed the strongest associations with septic shock, with shock occurring in 72.1% and 64.4% of affected patients, respectively. These associations remained statistically significant after adjustment for other variables in the multivariate logistic regression model. These findings are consistent with the pathophysiology of septic shock, in which an excessive inflammatory response leads to microcirculatory disturbances, impaired tissue perfusion, and multiple organ dysfunction. The cardiovascular and nervous systems are among the earliest and most severely affected organ systems in this process (Schlappbach et al., 2024; Weiss et al., 2020). The results of the current study are also in agreement with those reported by Trần et al. (2025), who found that fever (75.0%) and respiratory dysfunction (73.3%) were the most common clinical manifestations in pediatric sepsis, followed by cardiovascular dysfunction (45.0%), neurological dysfunction (36.7%), and gastrointestinal dysfunction (31.7%). This consistency suggests that fever and organ dysfunction are common clinical features of pediatric sepsis, reflecting the systemic inflammatory response and multiple organ injury caused by severe infection (Weiss et al., 2020).

In the present study, fever $>38.5^{\circ}\text{C}$ was identified as an independent predictor of septic shock, increasing the risk by 3.91-fold (adjusted odds ratio, 3.91; 95% confidence interval, 1.62–9.43; $p = 0.002$). Although respiratory and gastrointestinal dysfunction were associated with septic shock in univariate analyses, these associations were no longer statistically significant after adjustment for other variables in the multivariate model. This finding suggests that their effects may be outweighed by more severe organ dysfunction, particularly cardiovascular and neurological impairment. Taken together, these findings highlight the importance of clinical manifestations, especially high fever, cardiovascular dysfunction, and neurological dysfunction, in the early identification of children with sepsis who are at increased risk of developing septic shock. Close monitoring of these clinical signs from the time of hospital admission may assist clinicians in assessing disease severity, predicting clinical outcomes, and implementing timely therapeutic interventions to improve patient outcomes. Traditionally, blood culture has been considered the gold standard for the diagnosis of sepsis because of its ability to identify causative pathogens and determine antimicrobial susceptibility. However, this method has several limitations, including a prolonged turnaround time and a relatively low positivity rate, particularly in patients who have received antibiotics before hospital admission (Lamy et al., 2016; Trần et al., 2025). Therefore, biomarkers such as C-reactive protein, procalcitonin, and blood lactate have been increasingly used for the early diagnosis and prognostic assessment of sepsis and septic shock. Previous studies have demonstrated that procalcitonin has high sensitivity and specificity for the diagnosis of sepsis (Lê et al., 2022; Simon et al., 2004), whereas both C-reactive protein and procalcitonin have shown good discriminatory performance in septic patients with pronounced inflammatory responses, especially when multiple biomarkers are combined (Downes et al., 2020; Pierrakos et al., 2020; Zeng et al., 2022). Blood lactate, on the other hand, has been recognized as an indicator of tissue hypoperfusion and has been associated with severe disease progression and an increased risk of septic shock (Fernando et al., 2018; Jat et al., 2011).

In the present study, blood lactate was the biomarker with the best predictive performance for septic shock, with an AUC of 0.713 (95% confidence interval, 0.640–0.785; $p < 0.001$). The optimal cutoff value was 3.93 mmol/L,

yielding a high specificity of 87.1%. This finding is consistent with the study by Trần et al. (2025), which included 60 children with sepsis and reported that lactate was the only biomarker with significant prognostic value for septic shock, with an AUC of 0.791, a cutoff value of 5.09 mmol/L, a sensitivity of 66.7%, and a specificity of 85.7%. Similarly, Ma et al. (2023) found that blood lactate predicted septic shock with an AUC of 0.745, a sensitivity of 75%, and a specificity of 71%. Suwanpakdee et al. (2021) also demonstrated that initial lactate levels had excellent performance in the early detection of septic shock, with an AUC of 0.90, a sensitivity of 80%, and a specificity of 92.1%. Taken together, these findings indicate that blood lactate is a reliable biomarker reflecting tissue hypoperfusion and metabolic disturbances in patients with septic shock and therefore plays an important role in predicting disease severity and clinical outcomes.

Procalcitonin demonstrated a moderate predictive value for septic shock in the current study, with an AUC of 0.652 (95% confidence interval, 0.577–0.727; $p < 0.001$). The optimal cutoff value was 10.05 ng/mL, corresponding to a sensitivity of 52.4% and a specificity of 73.0%. This predictive performance was lower than that reported by Bustos & Padilla (2015) in Chile, where procalcitonin was identified as the best biomarker for predicting septic shock, with an AUC of 0.91, a sensitivity of 97%, and a specificity of 74%. Similarly, Carrol et al. (2005) reported that procalcitonin reliably discriminated between children with and without septic shock, with an AUC of 0.85, further supporting its potential value as a prognostic marker in pediatric sepsis. Xu et al. (2019) also found that procalcitonin had superior predictive performance compared with C-reactive protein, with AUC values of 0.78 and 0.65, respectively. The discrepancies among studies may be attributable to differences in study populations, diagnostic criteria, timing of biomarker measurements, and the proportion of patients with septic shock.

In contrast, C-reactive protein did not show significant predictive value for septic shock in the current study, with an AUC of 0.485 (95% confidence interval, 0.407–0.564; $p = 0.711$). This finding is consistent with the study by Trần et al. (2025), in which C-reactive protein yielded an AUC of only 0.505 and was not statistically significant in predicting septic shock. However, other studies have reported that C-reactive protein may have some value in the diagnosis or prognostic assessment of sepsis. Bustos & Padilla (2015) reported an AUC of 0.67 for C-reactive protein in predicting septic shock, whereas Zeng et al. (2022) found that the AUC of C-reactive protein could be as high as 0.974 in septic patients with hyperinflammatory states. These inconsistent findings suggest that the predictive performance of C-reactive protein is influenced by several factors, including the timing of disease onset, the intensity of the inflammatory response, and differences in study populations. Consequently, C-reactive protein generally exhibits lower predictive accuracy for septic shock than blood lactate and procalcitonin. Overall, the findings of the current study are consistent with the general trend reported in both domestic and international studies, demonstrating that blood lactate is the most effective biomarker for predicting septic shock, followed by procalcitonin, whereas C-reactive protein has limited predictive value. These results suggest that blood lactate and procalcitonin should be prioritized for the early assessment of septic shock risk in children with sepsis, particularly in healthcare settings with limited resources, such as hospitals in mountainous and remote areas.

Several limitations should be considered when interpreting the findings of this study. The retrospective design limited the completeness and consistency of the available clinical and laboratory data, while the inclusion of only patients with complete measurements of C-reactive protein, procalcitonin, and blood lactate may have introduced selection bias. Although the first available biomarker measurements obtained within 24 hours after hospital admission were used for analysis, the exact timing of sample collection varied among patients. Given the rapid clinical progression of pediatric sepsis, this variation may have affected biomarker concentrations and, consequently, their estimated predictive performance. The multicenter setting may also have introduced additional analytical variability because the participating hospitals did not necessarily use identical analyzers, assay platforms, reagents, or laboratory procedures, despite following standardized laboratory procedures and routine quality-control requirements. Furthermore, all participating hospitals were located in the northern mountainous provinces of Vietnam, where patient characteristics and healthcare accessibility may differ from those in other regions, which may limit the generalizability of the findings. Accordingly, the cutoff values identified for blood lactate and procalcitonin should be regarded as preliminary estimates derived from the present cohort rather than definitive thresholds for clinical decision-making. Further prospective multicenter studies involving independent and geographically diverse pediatric populations, with more standardized sampling times and laboratory methods, are needed to validate these findings and determine whether the proposed cutoff values can be reliably applied in routine clinical practice.

5. Conclusions

This multicenter retrospective study of 245 children with sepsis in the northern mountainous provinces of Vietnam demonstrated that the prevalence of septic shock was 33.5%. Clinical manifestations, including fever $>38.5^{\circ}\text{C}$, cardiovascular dysfunction, and neurological dysfunction, were identified as independent factors associated with septic shock, with cardiovascular dysfunction showing the strongest association. Among the three biomarkers evaluated, blood lactate showed the best predictive performance for septic shock, with an AUC of

0.713, followed by procalcitonin with an AUC of 0.652, whereas C-reactive protein did not demonstrate statistically significant predictive value. These findings suggest that blood lactate and procalcitonin may serve as useful tools for the early assessment of septic shock risk in children with sepsis, particularly in healthcare settings with limited resources, such as mountainous and remote areas. This study provides additional evidence regarding the prognostic value of biomarkers for septic shock in Vietnamese children and highlights the importance of integrating biomarkers with clinical manifestations to improve early detection, risk stratification, and optimization of treatment strategies for pediatric patients with sepsis.

Author Contributions

Conceptualization, D.T.K.T. and T.T.N.; methodology, D.T.K.T., T.T.N., and Đ.D.L.; validation, T.T.N. and C.T.N.; formal analysis, D.T.K.T. and T.T.N.; investigation, D.T.K.T., C.T.N., and Đ.D.L.; resources, C.T.N. and Đ.D.L.; data curation, D.T.K.T. and C.T.N.; writing—original draft preparation, D.T.K.T.; writing—review and editing, T.T.N., C.T.N., and Đ.D.L.; visualization, D.T.K.T.; supervision, T.T.N. and Đ.D.L.; project administration, T.T.N. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Written informed consent was obtained from all participants.

Ethical Approval

The study was approved by the Ethics Committee of Thai Nguyen University of Medicine and Pharmacy (Approval No. 125/ĐDD-BVYD; approved on 15 March 2025). All procedures involving human participants were conducted in accordance with the ethical standards of the institutional ethics committee and the principles of the Declaration of Helsinki. Written informed consent was obtained from all participants prior to their participation in the study.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to assist with English language editing and to improve the clarity and readability of the text. The AI tool was not used to generate, analyze, or interpret research data, nor to create scientific conclusions or references. The authors carefully reviewed and edited all AI-assisted content and take full responsibility for the accuracy, originality, and integrity of the manuscript.

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